A YOUNG DIAGRAM CAN BE TILED BY DOMINOES IF AND ONLY IF ITS ODD PARTS OCCUPY AS MANY EVEN AS ODD POSITIONS: THE CONJECTURE IN OEIS A300789

MATH0

Abstract. We prove that the Young diagram of an integer partition $y_1 \geq y_2 \geq \cdots \geq y_m$ can be tiled by dominoes if and only if the number of odd parts y_i with i even equals the number of odd parts y_i with i odd. Equivalently, the sequence A300789 of Heinz numbers of domino-tileable partitions coincides with the sequence of Heinz numbers of partitions satisfying this parity condition, as conjectured in the OEIS entry A300789 (G. Wiseman, 2018). The result is not new: it is the case $r = 2$ of the classical description of partitions with empty r-core, and the parity form appears explicitly in a lemma of Albion (arXiv 2021, published 2024); the OEIS entry nevertheless still records it as a conjecture. Our contribution is, to our knowledge, the first formal proof: it is verified in Lean 4, with Mathlib, against a statement that math0 formalized from the OEIS entry, reviewed for fidelity and frozen before any proof attempt, and whose definitions are checked by the kernel against the 63 listed terms.

1. Introduction

The Heinz number of an integer partition $y = (y_1, \ldots, y_m)$ is $p_{y_1} p_{y_2} \cdots p_{y_m}$, where p_j is the j-th prime ($p_1 = 2$); by unique factorization, every integer $k \geq 1$ is the Heinz number of exactly one partition, and $k = 1$ belongs to the empty partition. The OEIS entry A300789 [10], created by Gus Wiseman on March 12, 2018, lists the Heinz numbers of the partitions whose Young diagram can be tiled by dominoes:

$$1, 3, 4, 7, 9, 10, 12, 13, 16, 19, 21, 22, 25, 27, 28, 29, 34, 36, \ldots$$

The entry records the conjecture that these are exactly the Heinz numbers of the partitions "in which the odd parts appear as many times in even as in odd positions" (quoted in full in Section 2). For example, $10 = p_3 p_1$ is a term: $(3, 1)$ has odd parts in positions 1 and 2, and its diagram is tiled by one vertical and one horizontal domino. The odd parts of $(3, 2, 1)$, with Heinz number 30, are both in odd positions, and its diagram has four cells of one chessboard colour and two of the other. We prove the conjecture (Theorem 1 and Corollary 2).

Prior work and status. The statement is classical, and we do not claim it as new. The necessity of the condition is the chessboard-colouring argument: each domino covers one cell of each colour, and a row of length y_i contributes $\pm[y_i \text{ odd}]$ to the colour difference (Lemma 3). For sufficiency, in the language of cores, a partition's diagram is domino-tileable exactly when its 2-core is empty: removing rim dominoes, which preserves the colour difference, leads to the 2-core, which is a staircase $(k, k-1, \ldots, 1)$, and a nonempty staircase has nonzero colour difference. Littlewood's decomposition of a partition into its r-core and r-quotient [5] is classical; see James and Kerber [4] and Macdonald [6, Ch. I, §1, Ex. 8, pp. 12–15], where the r-core is read off from the distribution of the beta-numbers of the partition modulo r; when the number of beta-numbers is a multiple of r, the r-core is empty exactly when they are equidistributed modulo r. For $r = 2$ this is the parity condition in another form. Albion [1, Lemma 2.1], who

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takes domino-tileability as a definition of empty 2-core, states for λ with at most $2n$ parts that λ has empty 2-core if and only if $\{\lambda_i + 2n - i : 1 \le i \le 2n\}$ contains n even and n odd integers, noting that the equivalences "easily follow by induction on $|\lambda|$". Since $\lambda_i + 2n - i \equiv \lambda_i + i \pmod 2$, counting the even and odd values position by position shows that this is exactly the condition of A300789. The black-equals-white criterion for Young diagrams is also the key fact behind the competitive-programming problem "Domino for Young" (Codeforces Round #609, December 2019, problems 1268B and 1269D [2]), whose answer, the maximum number of disjoint dominoes in a Young diagram, is the smaller of the two colour counts.

Within the OEIS, the entry A304710 [11] (A. P. Heinz, May 17, 2018), which counts the partitions of $2n$ whose diagram cannot be tiled with dominoes, states, without the word "conjectured", that it also counts the partitions of $2n$ in which the number of odd parts in even positions differs from the number in odd positions, and its example section gives the colouring argument for $(3, 2, 1)$. In A000712 [9] (partitions of n into parts of two kinds), W. Meeussen (April 17, 2013) and J. Karlsson (December 9, 2025) commented that it also counts the partitions of $2n$ satisfying the parity condition, respectively the domino-tileable ones. Both programs in A300789 itself generate the terms from the parity condition: the Mathematica program is labelled "Conjectured", while the Maple program (A. P. Heinz, May 22, 2018) carries no such label. The entry (revision #11, May 22, 2018) still describes the statement as a conjecture and cites no proof; the statement is not among those of the formal-conjectures repository [3]. We therefore classify our result as the first formal proof, to our knowledge, of a known result.

Method. We follow the standard argument: the chessboard imbalance of a diagram vanishes exactly under the parity condition, and it vanishes for every tileable diagram (Lemmas 4 and 6); conversely, a balanced diagram is tiled by removing rim dominoes, since a diagram without a removable rim domino is a staircase (Proposition 8).

2. Statement

We index rows and columns from 0. For a partition $y = (y_1 \ge \cdots \ge y_m)$ with $y_m \ge 1$ let $Y(i) = y_{i+1}$ for $i < m$ and $Y(i) = 0$ for $i \ge m$. The *Young diagram* of y is $\{(i, j) \in \mathbb{N}^2 : j < Y(i)\}$. A *domino* is a set $\{(i, j), (i+1, j)\}$ or $\{(i, j), (i, j+1)\}$, and a set $S \subseteq \mathbb{N}^2$ is *domino-tileable* if it is the union of a family of pairwise disjoint dominoes. Positions are numbered $1, \ldots, m$ in the order $y_1, \ldots, y_m$.

Theorem 1. *The Young diagram of a partition $y = (y_1 \ge \cdots \ge y_m)$ is domino-tileable if and only if*

$$\#\{1 \le i \le m : i \text{ even},\ y_i \text{ odd}\} = \#\{1 \le i \le m : i \text{ odd},\ y_i \text{ odd}\}. \tag{1}$$

Reversing the order of the parts maps position i to $m+1-i$, which either preserves all parities or swaps all of them; hence (1) does not depend on whether the parts are listed in decreasing or increasing order (the entry's programs use the increasing order).

Corollary 2. *Let $a(n)$ be the n-th smallest Heinz number of a partition with domino-tileable Young diagram, and $b(n)$ the n-th smallest Heinz number of a partition satisfying* (1)*. Then $a(n) = b(n)$ for all $n \ge 1$.*

The entry A300789 [10] has the name "Heinz numbers of integer partitions whose Young diagram can be tiled by dominos." and offset 1; its second comment reads, verbatim:

> This sequence is conjectured to be the Heinz numbers of integer partitions in which the odd parts appear as many times in even as in odd positions.

The comment is unsigned; the entry's author is Gus Wiseman (March 12, 2018), and the last revision (#11) is dated May 22, 2018.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A300789.lean`, reviewed for fidelity and frozen before any proof attempt (SHA-256 `6d088323...05be`, recorded at the fidelity review and listed in `lean/Math0/Statements/FROZEN.sha256`; the proof leaves the file unchanged). Its definitions read as follows (verbatim; the import, docstrings, blank lines, the decision procedure, the decidability instances and all theorems are omitted):

```
namespace Math0.Statements.A300789
def parts (k : ℕ) : List ℕ := (k.primeFactorsList.map Nat.primeCounting).reverse
def youngDiagram (y : List ℕ) : Set (ℕ × ℕ) :=
  {c | c.1 < y.length ∧ c.2 < y.getD c.1 0}
def IsDomino (d : Set (ℕ × ℕ)) : Prop :=
  ∃ i j : ℕ, d = {(i, j), (i + 1, j)} ∨ d = {(i, j), (i, j + 1)}
def DominoTileable (S : Set (ℕ × ℕ)) : Prop :=
  ∃ D : Set (Set (ℕ × ℕ)), (∀ d ∈ D, IsDomino d) ∧ D.PairwiseDisjoint id ∧ ⋃₀ D = S
def IsTerm (k : ℕ) : Prop := 0 < k ∧ DominoTileable (youngDiagram (parts k))
noncomputable def a (n : ℕ) : ℕ := Nat.nth IsTerm (n - 1)
def OddPartsBalanced (y : List ℕ) : Prop :=
  ((Finset.range y.length).filter fun i => Even (i + 1) ∧ Odd (y.getD i 0)).card =
    ((Finset.range y.length).filter fun i => Odd (i + 1) ∧ Odd (y.getD i 0)).card
def IsConjTerm (k : ℕ) : Prop := 0 < k ∧ OddPartsBalanced (parts k)
noncomputable def b (n : ℕ) : ℕ := Nat.nth IsConjTerm (n - 1)
def statement : Prop := ∀ n : ℕ, 1 ≤ n → a n = b n
end Math0.Statements.A300789
```

Here `Nat.primeFactorsList k` is the increasing list of prime factors of k with multiplicity and `Nat.primeCounting p` $= \pi(p)$ is the index of the prime p, so `parts k` is the partition with Heinz number k in weakly decreasing order. As `y.getD i 0` is the i-th entry of `y` (from 0), or 0 beyond its length, `youngDiagram` is the diagram defined above and `OddPartsBalanced` is (1) with position $i+1$ for index i. `Nat.nth p j` is the j-th smallest (from 0) number satisfying p, so `a` and `b` are the sequences of Corollary 2 with offset 1. The statement file decides `IsTerm` by a verified search (Section 3.1) and proves by `decide +kernel`, i.e. by evaluation in the kernel, that $a(1),\dots,a(63)$ and $b(1),\dots,b(63)$ are both the 63 terms $1,3,\dots,131$ listed in the entry (`a_data`, `b_data`; a kernel-checked `List.filter` over $\{0,\dots,131\}$ is transferred to `Nat.nth` by `map_nth_eq_of_filter`), and that `parts` gives the partitions of the entry's example (`parts_example`). The conjecture is read as equality of the two increasing sequences, with no extra hypotheses; Corollary 2 is exactly `statement`. Both term sets contain every 3^e (the partition $(2,\dots,2)$), so they are infinite and this reading is equivalent to equality of the two sets; the proof establishes the latter directly. Since every partition is `parts k` for its Heinz number k, equality of the two sets is in turn equivalent to Theorem 1; this last step uses the bijection between partitions and Heinz numbers, which is not formalized (see the remark after the proof of Theorem 1).

3. Proof

The proof follows `lean/Math0/Frontier/OEIS/A300789.lean`; Lean names are given in parentheses. Write $[\,P\,] \in \{0,1\}$ for the truth value of P. For a function $Y\colon \mathbb{N}\to\mathbb{N}$ and $N\in\mathbb{N}$ define the *imbalance* and the *diagram*

$$\mathrm{I}_N(Y) = \sum_{i<N} (-1)^i\,[\,Y(i) \text{ odd}\,] \in \mathbb{Z} \quad (\texttt{imb}), \qquad \mathrm{D}(Y) = \{(i,j)\in\mathbb{N}^2 : j < Y(i)\} \quad (\texttt{shape}).$$

For a list y of length m we write $Y_y(i) =$ `y.getD i 0`, and give the cell $c=(i,j)$ the weight $w(c) = (-1)^{i+j}$ (`wt`).

Lemma 3 (`sum_row`). *For all i, n: $\sum_{j<n}(-1)^{i+j} = (-1)^i\,[\,n \text{ odd}\,]$.*

Proof. Induction on n: adding the term $(-1)^{i+n}$ changes $(-1)^i[\,n \text{ odd}\,]$ into $(-1)^i[\,n+1 \text{ odd}\,]$, as one checks separately for n even and n odd. □

Lemma 4 (`term_eq`, `balanced_iff`)**.** *For every list y of length m, `OddPartsBalanced y` holds if and only if $\mathrm{I}_m(Y_y) = 0$. In particular, for a partition y, condition* (1) *is equivalent to $\mathrm{I}_m(Y_y) = 0$.*

Proof. For $i, v \in \mathbb{N}$ we have $(-1)^i[\,v \text{ odd}\,] = [\,i+1 \text{ odd} \wedge v \text{ odd}\,] - [\,i+1 \text{ even} \wedge v \text{ odd}\,]$, by checking the four parity cases of i and v. Summing over $i < m$ with $v = Y_y(i)$ gives $\mathrm{I}_m(Y_y) = B - A$, where A and B are the two cardinalities in `OddPartsBalanced` (even and odd positions, respectively). □

3.1. **Tileable diagrams are balanced.** Rather than summing weights over an abstract family of dominoes, the Lean proof runs the colour argument along the decision procedure of the statement file. For a list y, `cellsList y` lists the cells of `youngDiagram y` row by row, in strictly increasing row-major order (`pairwise_cellsList`), and `tileL` tries to pair the first cell of a list with its right neighbour or with the cell below:

```
def tileL : ℕ → List (ℕ × ℕ) → Bool
  | _, [] => true
  | 0, _ :: _ => false
  | fuel + 1, c :: rest =>
    (decide ((c.1, c.2 + 1) ∈ rest) && tileL fuel (rest.erase (c.1, c.2 + 1))) ||
      (decide ((c.1 + 1, c.2) ∈ rest) && tileL fuel (rest.erase (c.1 + 1, c.2)))
```

Lemma 5 (`tileL_iff`, in the statement file)**.** *If the list l of cells is strictly increasing in row-major order and has length at most f, then `tileL f l = true` if and only if the set of entries of l is domino-tileable.*

Proof sketch. Induction on l. If S is tileable and c is its row-major first cell, the domino $d \ni c$ of a tiling lies in S, so its other cell is not before c; hence $d = \{c, c+(0,1)\}$ or $d = \{c, c+(1,0)\}$ (`domino_at_min`), and $S \setminus d$ is tiled by the remaining dominoes (`exists_domino_of_tileable`). Conversely, adding a domino $d \subseteq S$ to a tiling of $S \setminus d$ tiles S (`tileable_of_diff`). Removing two entries from a strictly increasing list leaves a strictly increasing list. □

Lemma 6 (`sum_wt_of_tileL`, `sum_cellsList`)**.** *If `tileL f l = true`, then $\sum_{c \in l} w(c) = 0$. For every list y of length m, $\sum_{c \in \mathit{cellsList}\ y} w(c) = \mathrm{I}_m(Y_y)$.*

Proof. For the first claim, induct on f. If $l = c :: \mathit{rest}$ is accepted, then some $c' \in \{c+(0,1), c+(1,0)\}$ lies in *rest* and *rest*.erase c' is accepted with fuel $f-1$, so its weight is 0 by induction. Since $c :: \mathit{rest}$ is a permutation of $c :: c' :: (\mathit{rest}.\mathrm{erase}\ c')$ and $w(c') = -w(c)$, the weight of l is $w(c) + w(c') + 0 = 0$. The second claim follows by summing Lemma 3 over the rows $i < m$. □

3.2. **Balanced diagrams are tileable.**

Lemma 7 (`imb_staircase_ne`)**.** *Let $N \in \mathbb{N}$ and $Y \colon \mathbb{N} \to \mathbb{N}$ with $Y(i) = 0$ for $i \geq N$, $k = Y(0) \geq 1$, and $Y(r) = \max(k-r, 0)$ for all r. Then $\mathrm{I}_N(Y) \neq 0$.*

Proof. From $0 = Y(N) = \max(k-N, 0)$ we get $k \leq N$. If $Y(i)$ is odd then $Y(i) = k - i$, so $i \equiv k+1 \pmod 2$ and $(-1)^i = (-1)^{k+1}$. Hence $\mathrm{I}_N(Y) = (-1)^{k+1} \sum_{i<N} [\,Y(i) \text{ odd}\,]$. The last sum is at least $[\,Y(k-1) \text{ odd}\,] = 1$, since $k - 1 < N$ and $Y(k-1) = 1$. □

Proposition 8 (`tileable_shape`)**.** *Let $N \in \mathbb{N}$ and let $Y \colon \mathbb{N} \to \mathbb{N}$ be weakly decreasing with $Y(i) = 0$ for $i \geq N$ and $\mathrm{I}_N(Y) = 0$. Then $\mathrm{D}(Y)$ is domino-tileable.*

Proof. By strong induction on $s = \sum_{i<N} Y(i)$, the number of cells, for all such Y simultaneously. If $Y(0) = 0$ then $Y \equiv 0$, $\mathrm{D}(Y) = \emptyset$, and the empty family tiles it. Assume $Y(0) \geq 1$. In the first two cases below we exhibit a domino $d \subseteq \mathrm{D}(Y)$ and a function Y' satisfying the hypotheses with $\sum_{i<N} Y'(i) < s$ and $\mathrm{D}(Y) \setminus d = \mathrm{D}(Y')$; by induction $\mathrm{D}(Y')$ is tileable, and adding d to its tiling tiles $\mathrm{D}(Y)$ (`tileable_of_diff`).

Case 1: $Y(i) \geq Y(i+1) + 2$ for some i. Then $Y(i) \geq 2$, so $i < N$. Let $d = \{(i, Y(i) - 2), (i, Y(i) - 1)\}$, a horizontal domino in $\mathrm{D}(Y)$, and let Y' agree with Y except $Y'(i) = Y(i) - 2$. Clearly $\mathrm{D}(Y) \setminus d = \mathrm{D}(Y')$. Y' is weakly decreasing because $Y'(i) = Y(i) - 2 \geq Y(i+1)$ and $Y'(i) < Y(i) \leq Y(a)$ for $a < i$, it vanishes from N on, it has two cells fewer, and $\mathrm{I}_N(Y') = \mathrm{I}_N(Y) = 0$ since $Y(i) - 2 \equiv Y(i) \pmod 2$.

Case 2: $Y(i) = Y(i+1) > Y(i+2)$ for some i. Then $Y(i+1) \geq 1$, so $i + 1 < N$. Let $d = \{(i, Y(i) - 1), (i+1, Y(i) - 1)\}$, a vertical domino in $\mathrm{D}(Y)$, and let Y' agree with Y except $Y'(i) = Y(i) - 1$ and $Y'(i+1) = Y(i+1) - 1$. Again $\mathrm{D}(Y) \setminus d = \mathrm{D}(Y')$. Y' is weakly decreasing: for $a < i$ we have $Y'(a) = Y(a) \geq Y(i) > Y'(i)$, and for $b \geq i + 2$ we have $Y'(b) = Y(b) \leq Y(i+2) \leq Y(i+1) - 1 = Y'(i+1) = Y'(i)$. The rows i and $i+1$ have equal lengths before and after, so their contributions $(-1)^i\varepsilon + (-1)^{i+1}\varepsilon$ to the imbalance (with $\varepsilon = [\,Y(i) \text{ odd}\,]$, respectively $[\,Y(i) - 1 \text{ odd}\,]$) are 0 in both Y and Y', while all other rows are unchanged; hence $\mathrm{I}_N(Y') = \mathrm{I}_N(Y) = 0$.

Case 3: neither case applies. Then for every i, $Y(i+1) \leq Y(i) \leq Y(i+1) + 1$, and $Y(i+1) = Y(i)$ implies $Y(i+2) \geq Y(i+1)$, hence $Y(i+2) = Y(i+1)$. So if $Y(r+1) = Y(r)$ for some r, then by induction $Y(r+t+1) = Y(r+t)$ for all t, and $Y(r) = Y(r+N) = 0$ (`hflat`). We now show $Y(r) = \max(k - r, 0)$ with $k = Y(0)$, by induction on r (`hst`). The case $r = 0$ is clear. Given it for r: either $Y(r+1) = Y(r)$, and then $Y(r) = 0$, so $k \leq r$ and $Y(r+1) = 0 = \max(k - r - 1, 0)$; or $Y(r+1) = Y(r) - 1$, and then $Y(r) \geq 1$, so $Y(r) = k - r$ and $Y(r+1) = k - (r+1)$. Thus Y is a staircase with $Y(0) \geq 1$, and Lemma 7 gives $\mathrm{I}_N(Y) \neq 0$, a contradiction. So Case 3 cannot occur. □

In Lean, Y' is the function `fun r => if r = i then Y r - 2 else Y r` (in Case 2, `if r = i ∨ r = i + 1 then Y r - 1 else Y r`), so no list manipulation is needed; the arithmetic side conditions are discharged by `omega`.

3.3. **Proof of Theorem 1 and Corollary 2.**

Proof of Theorem 1. Let y be a partition with m parts, so that Y_y is weakly decreasing, vanishes from m on, and `youngDiagram y` $= \mathrm{D}(Y_y)$ (`youngDiagram_eq`; the condition $i < m$ in `youngDiagram` is implied by $j < Y_y(i)$). By Lemma 4, (1) is equivalent to $\mathrm{I}_m(Y_y) = 0$. If the diagram is tileable, Lemma 5 (with $l =$ `cellsList` y and f its length) shows that `tileL` accepts `cellsList y`, and Lemma 6 gives $\mathrm{I}_m(Y_y) = 0$. Conversely, if $\mathrm{I}_m(Y_y) = 0$, Proposition 8 with $N = m$ shows that $\mathrm{D}(Y_y)$ is tileable. □

In Lean the theorem is proved as `isTerm_iff`: `IsTerm k` $\leftrightarrow$ `IsConjTerm k` for every $k \in \mathbb{N}$, by this argument for $y =$ `parts k`, which is weakly decreasing because `Nat.primeFactorsList k` is sorted and π is monotone (`parts_pairwise`, `antitone_getD`). All other lemmas hold for arbitrary lists, respectively weakly decreasing functions; the bijection between partitions and Heinz numbers is not needed for the formal statement and is not formalized.

Proof of Corollary 2. By `isTerm_iff` the predicates `IsTerm` and `IsConjTerm` are equal (function and propositional extensionality), hence so are `Nat.nth IsTerm` and `Nat.nth IsConjTerm`, and `a n = b n` for every n, in particular for $n \geq 1$ (`result`). □

4. Machine verification

Files and declarations. The proof is the file `lean/Math0/Frontier/OEIS/A300789.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A300789.lean` and the audit commands of `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A300789.result`, of type `Math0.Statements.A300789.statement`; the auxiliary declarations of Section 3 are private. From the statement file the proof uses `tileL_iff`, `pairwise_cellsList`, `setOf_mem_cellsList`, `tileable_of_diff` and `tileable_empty`. The file ends with the command `#math0_settles Math0.Frontier.A300789.result Math0.Statements.A300789.statement`. The proof file contains no `sorry`, `native_decide`, `axiom` or `set_option`, and the statement file's data lemmas use only `decide +kernel`.

Audit. The command `#math0_settles` (`lean/Math0/Audit.lean`) fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (rejecting `sorryAx`, the axioms of `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or its negation, up to metadata. Here it reports

```
math0 audit: 'Math0.Statements.A300789.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A300789.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A300789.result' proves 'Math0.Statements.A300789.statement'
```

Toolchain and reproduction. We use Lean 4 [8] `v4.33.1` and Mathlib [7] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [3] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A300789.lean` in the directory `lean`.

References

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[2] Codeforces Round #609, Problem 1268B (Div. 1) / 1269D (Div. 2), *Domino for Young*, December 2019, `https://codeforces.com/problemset/problem/1268/B`; editorial at `https://codeforces.com/blog/entry/72358`.

[3] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[4] G. James and A. Kerber, *The Representation Theory of the Symmetric Group*, Encyclopedia of Mathematics and its Applications **16**, Addison-Wesley, Reading, MA, 1981.

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[10] The OEIS Foundation Inc., `https://oeis.org/A300789`, entry by G. Wiseman (Mar. 12, 2018); revision #11, May 22, 2018.
[11] The OEIS Foundation Inc., `https://oeis.org/A304710`, entry by A. P. Heinz (May 17, 2018).

Produced by Claude (an AI model developed by Anthropic) in the math0 project